

Zackory Erickson

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Carnegie Mellon University
Robotics Institute
5000 Forbes Avenue
Pittsburgh, PA 15213

Current Positions

Carnegie Mellon University , Robotics Institute	
Raj Reddy Assistant Professor	2026–present
Assistant Professor	2021–2026
Courtesy appointment: Biomedical Engineering	
Founding Scientific Advisor , Intuitive Autonomy	2025–present
Advisor , Skylark Labs	2026–present

Education

Ph.D., Robotics	2016–2021
Georgia Institute of Technology	
Advisor: Charles C. Kemp	
M.S., Computer Science	2020
Georgia Institute of Technology	
Advisor: Charles C. Kemp	
B.S., Computer Science, Mathematics (double major)	2012–2016
University of Wisconsin–La Crosse	

Honors and Awards

Raj Reddy Career Development Professorship in Robotics	2026
NSF CAREER	2026
<i>“CAREER: Scaling Up Physical Human-Robot Interaction with Generative Simulation”</i>	
Best Paper Award , ACM/IEEE International Conference on Human-Robot Interaction (HRI)	2026
<i>“WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding”</i>	
Best Paper Finalist , ACM/IEEE International Conference on Human-Robot Interaction (HRI)	2026
<i>“Bidirectional Human-Robot Communication for Physical Human-Robot Interaction”</i>	
Best Paper Finalist , ACM/IEEE International Conference on Human-Robot Interaction (HRI)	2025
<i>“LAMS: LLM-Driven Automatic Mode Switching for Assistive Teleoperation”</i>	
OSCAR Scholar , University of Pittsburgh, Alzheimer’s Disease Research Center (ADRC)	2024
Best Paper Award , ACM/IEEE International Conference on Human-Robot Interaction (HRI)	2024
<i>“Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator”</i>	
Best Poster Award , CMU Mechanical Engineering Research Symposium	2024
<i>“Extended Kalman Filter for Real-Time State Estimation of Cloth for Robot Manipulation”</i>	
NVIDIA Academic Hardware Grant	2022
<i>“Adapting to Distribution Shift in Deformable Manipulation with Assistive Robots”</i>	

NVIDIA Fellowship Finalist	2020
Best Student Paper Award , IEEE International Conference on Rehabilitation Robotics (ICORR) <i>“Multidimensional Capacitive Sensing for Robot-Assisted Dressing and Bathing”</i>	2019
Best Paper Award in Service Robotics Finalist , IEEE ICRA <i>“Classification of Household Materials via Spectroscopy”</i>	2019
President’s Fellowship , Georgia Tech	2016–2020
4th Heidelberg Laureate Forum	2016
Honorable Mention , NSF GRFP	2016
Strzelczyk Award Awarded to the top graduating senior in the College of Science and Health for academic achievement and service to the campus and community.	2016
MIT CONVERGE One of 18 prospective PhD students in the nation invited to tour MIT.	2015
Berkeley Engineering Preview Days One of 14 prospective PhD students nationwide invited to tour UC Berkeley.	2015
Grace Olwell Memorial Endowment Fund Scholarship	2015
Xcel Energy Scholarship	2015
John and Lois Storlie Scholarship in Computer Science	2014
Undergraduate Research Grant , UW–La Crosse	2013
Scottish Rite Abbott Scholarship	2013
Dean’s List , UW–La Crosse	8 semesters

Mentoring

PhD Students

Cailyn Smith , CMU, RI	2025–
Yucheng Li , CMU, Heinz	2025–
Junxiang (Jim) Wang , CMU, RI	2024–
Zheyuan Hu , CMU, RI (co-advised with Aviral Kumar)	2024–
Hongyi Chen , CMU, RI (co-advised with Jeffrey Ichnowski)	2024–
Kavya Puthuveetil , CMU, RI (co-advised with Reid Simmons)	2022–
Zulekha Karachiwalla , CMU, RI (co-advised with Henny Admoni)	2022–2025
Je-Han Yang , CMU, BME (co-advised with Douglas Weber)	2022–
Yufei Wang , CMU, RI (co-advised with David Held)	2021–
Akhil Padmanabha , CMU, RI (co-advised with Carmel Majidi) Current: Meta	2021–2025

M.S. Students

Niharika Pant , CMU, MechE	2025–
Yitong (Elly) Wang , CMU, MLD	2025–
Kshitiz Kshitiz , CMU, RI	2025–
Saksham Bhutani , CMU, RI	2025–
Shikun Ban , CMU, RI	2025–
Shohei Nagai , CMU, RI	2025–
Tianhui (Tiffany) Han , CMU, RI	2025–
Anjana Ramachandran , CMU, BME-AIE	2025–
Julian Millan , CMU, MechE	2025–
Mengning Wu , CMU, RI	2025–
Qihao Lin , CMU, BME	2025–
Yinong He , CMU, RI	2025–
Crystal Zhou , CMU, MechE	2025–
MRSD Project: Operating room logistics 2.0 , CMU, RI	2025–
Students: Ayush Sharan, Saketh Polamraju, Akshit Raizada, Mehar Goenka, Teja Kanthamneni	
Hanqi Feng , CMU, RI	2024–
Alexis Hao , CMU, RI	2024–
Tiancheng (Tony) Wu , CMU, RI	2024–
Mino Nakura , CMU, RI	2024–
Pratik Bhowal , CMU, RI	2024–
Divyam Goel , CMU, RI	2024–
Xinwen Xu , CMU, MechE	2024–
Aditi Guruprasad , University of Pittsburgh, Bioengineering	2024–2025
Jing Gao , CMU, RI	2024–2025
Zoe LaLena , CMU, RI	2024–2025
Yatharth Ahuja , CMU, RI	2023–2024
Yiran Tao , CMU, RI	2023–2025
Current: PhD Student at MIT	
Michaela Tecson , CMU, RI	2023–
Tariq Hussain , CMU, MechE	2023–2025
Yexin Hu , CMU, CEE	2023–2025
Current: PhD Student at TU Wien	
Abhishek Tandon , CMU, RI	2023–2024
Current: Adobe	
Anujraaj (Argo) Goyal , CMU, RI	2023–2024
Current: Snapchat	
Zhanyi Sun , CMU, RI	2023–2024

Current: PhD Student at Stanford

MRSD Project: Operating room logistics , CMU, RI	2023–2025
Students: Gaurav Sethia, Jinkai Qiu, Roman Kaufman, Tanmay Agarwal, Yungjun Kim, Siddharth Ghodasara	
Sankalp Chopkar , CMU, RI	2022–2024
Atharva Kusalkar , CMU, RI	2022–2024
Saurav Kambil , CMU, MechE	2022–2024
Zilin Zhang , CMU, RI	2022–2024
MRSD Project: Auxilio , CMU, RI	2022–2023
Students: Shaolin Kataria, Shivam Tripathy, Praveen Venkatesh, Abhinav Gupta, Atharva Pusalkar	
Fukang Liu , CMU, MechE	2021–2023
Current: PhD Student at Georgia Tech	
Vaidehi Patil , CMU, RI	2021–2022
MRSD Project: TouRI , CMU, RI	2021–2022
Students: Shivani Sivakumar, Jashkumar Diyora, Shruti Gangopadhyay, Prakhar Pradeep, Jigarkumar Patel	
Pratyusha Karnati , Georgia Tech, CS	2020–2021
Current: Google X Robotics	
Yijun (Esther) Gu , Georgia Tech, CS	2019–2021
Current: PhD Student at Imperial College London	

Undergraduate Students

Nir Pechuk , CMU, CS	2026–
Naveen Enock , CMU, CIT	2025–
Jasmine Li , CMU, ECE	2025–
Robyn Wu , CMU, CS	2025–
Riya Kadakia , CMU, CS	2025–
Karolena Zhou , CMU, CIT	2024–
Divya Gupta , CMU, CS	2024–
Lehong Wu , Peking University, CS	2024–
Vivianna Lieu , CMU, CS	2024–
Annika Srinivasan , UIUC, MechE	2024–2025
Cindy Sun , CMU, MechE	2024–
Victoria de Leon , ITESM (NSF REU, RISS), Robotics	2024–2025
Tanisha Mehta , CMU, ECE, CS	2024–2025
Catherine Li , CMU, CS	2024–
Navin Sriram , Indian Institute of Technology Madras, CS	2024–2025
Sreyas Venkataraman , Indian Institute of Technology, Kharagpur, CS	2024–2025
Ziyu Wang , Tsinghua University, IIIS	2024–2025

Bharath Hegde , BITS Pilani University, CS	2024–2025
Zhengkao (Cary) Jin , CMU, ECE	2024–2025
Glenda Tan , CMU, CS	2024–
Jessica Han , CMU, CS	2024–
Ryan Ding , CMU, CS	2024–
Pranavi Kondapalli , CMU, CS	2024–
Maxwell Soh , CMU, CS	2023–2024
Ellen Lee , CMU, CS	2023–2024
Weikang Wan , Peking University, CS	2023–2024
Chen Chen , Tsinghua University, CS	2023–2024
Janavi Gupta , CMU, CS	2023–
Sumayya Syeda , CMU, CIT	2023–2024
Preethi Krishnamoorthy , CMU, CS	2023–2024
Jessie Yuan , CMU, CS	2023–2024
Madeliene Brutin , CMU, CS	2023–2024
Kendra Givens , Middle Tennessee State University (NSF REU, RISS), CS Current: PhD student at Vanderbilt University	2023–2024
Alexandra Gillespie , Colby College (NSF REU, RISS), CS Current: PhD student at University of Edinburgh	2023–2024
Yikang (Bruce) Cheng , CMU, CS	2022–2023
Jacob Delgado-López , University of Puerto Rico (NSF REU, RISS), CS Current: PhD student at JHU	2022
Wesley Lewis , University of Virginia (NSF REU, RISS), CS	2022–2024
Allen Zheng , CMU, CS	2022–2024
Qin (Alicia) Wang , CMU, CS	2022–2023
Daphne Han , CMU, CB	2022
Alexandra (Sasha) Wald , CMU, CS	2021–2024
Kavya Puthuveetil , Virginia Commonwealth University (NSF REU), BME	2021–2022
Samantha Mutiti , Georgia Tech, BME	2021
Holden Schaffer , Georgia Tech, CS	2020–2021
Siyan (Sylvia) Li , Georgia Tech, CS	2018–2019
Jiaqi (Julia) Chen , Georgia Tech, CS Current: PhD Student at ETH Zurich	2018
Katelyn Sosnowski , University of Arizona (NSF REU), BME Current: BME PhD Student at University of Arizona	2018
Mallak Taleb , University of Michigan (NSF REU), BME	2018
Bharat Srirangam , Georgia Tech, CS Current: Woot, Inc.	2018–2020

Eliot Xing , Georgia Tech, CE	2017–2022
Vamsee Gangaram , Georgia Tech, CS Current: Microsoft	2017–2020
Jong Hwa (Austin) Jang , Georgia Tech, CS	2017–2018
Maggie Collier , University of Alabama at Birmingham (NSF REU), BME Current: Robotics PhD student at CMU, NDSEG fellow	2017
Nathan Luskey , Georgia Tech, BME Current: MSCS student at CMU	2017–2018

Teaching

16-467: Introduction to Human-Robot Interaction , CMU	Spring 2025
16-762: Mobile Manipulation , CMU	Spring 2024, 2026–
16-741: Mechanics of Manipulation , CMU	Fall 2022–
16-887: Robotic Caregivers and Intelligent Physical Collaboration , CMU	Spring 2022–2023
Robotic Caregivers (BMED 4833/8813) , Co-instructor, Georgia Tech	Spring 2021
Robotic Caregivers (BMED 4803/8813) , Co-developer and instructor, Georgia Tech	Spring 2020

Invited Talks

<i>Robotic Caregivers and Wearable Health Sensors</i> Samsung Research America Allegheny Health Network Shanghai Jiao Tong University School of Medicine	2025 2025 2025
<i>Physical HRI with Deformable Manipulation</i> ICRA 2025 Workshop on Reflections on Representations and Manipulating Deformable Objects	2025
<i>Robot Learning and Wearable Interfaces in Pursuit of Robotic Caregivers</i> University of Wisconsin–Madison Nankai University National University of Singapore University of Texas at Austin	2025 2025 2025 2024
<i>Robot Learning, Wearable Sensing, and Teleoperation in Pursuit of Robotic Caregivers</i> 16-761: Mobile Robotics, CMU CMU RI Seminar RISS RoboLaunch Northeastern University University of Utah University of Illinois Urbana-Champaign	2024 2023 2023 2023 2023 2023
<i>My Journey to Professor</i> Robotics Institute Summer Scholars Seminar, CMU	2023
<i>Robotics and Artificial Intelligence in Healthcare and Opportunities for Innovation</i> Clinical Impact Symposia, BrightSpring Health Services	2023
<i>Capacitive Servoing and Spectroscopy for Physically Assistive Robotics</i> 16-722: Sensing and Sensors, CMU	2022
<i>Haptic Perspective-taking from Vision and Force</i>	

CMU RI Seminar	2022
<i>Capacitive Proximity Servoing for Physically Assistive Robotics</i>	
RSS 2022 Workshop on Advances in Close Proximity Human-Robot Collaboration	2022
<i>Robotic Caregiving and Human Interaction</i>	
24-675: Humanoid Robotics and Cognition, CMU	2022
16-311: Introduction to Robotics, CMU	2022
05-899: Special Topics in HCI: Accessibility, CMU	2021
<i>Capacitive Servoing for Physically Assistive Robotics</i>	
IROS 2021 4th Workshop on Proximity Perception in Robotics	2021
<i>Robotic Caregivers—Sensing, Simulation, and Physical Human-Robot Interaction</i>	
Carnegie Mellon University	2021
University of Pennsylvania	2021
<i>Physics-based Cloth Simulation and Learning Towards Robotic Caregiving</i>	
ICRA 2021 Workshop on Representing and Manipulating Deformable Objects	2021
<i>Robotic Caregivers—Recent Advances in Physics-based Simulation</i>	
Medical Robotics Club, Georgia Tech	2021
<i>Robot-Assisted Dressing</i>	
RESNA 2018 Workshop on Smart and Robotic Homes	2018
<i>Multimodal Anomaly Detection</i>	
Mathematics Colloquium, UW–La Crosse	2015

Academic Service

Chair of PhD Admissions Committee — CMU, Robotics Institute	2025–
Workshop Organizer — IROS, Robotic MANipulation of Deformable Objects (ROMADO)	2025
Area Chair — Robotics: Science and Systems (RSS)	2025
Workshop Organizer — HRI, Enhancing Independence with Physical Caregiving Robots	2025
Co-Chair of PhD Admissions Committee — CMU, Robotics Institute	2024
Workshop Organizer — RSS, Learning for Assistive Robotics	2024
Faculty Senate — CMU, representing the Robotics Institute	2024–
Advisory board — University of Wisconsin–La Crosse, Computer Science Department	2024–
Panelist — HRI, Workshop on HRI for Aging in Place	2024
Workshop Organizer — ICRA, Exploring Role Allocation in Human-Robot Co-Manipulation	2024
Associate Editor — IEEE Robotics and Automation Letters (RA-L) Manipulation and Grasping	2023–
NSF AI-CARING Arena Lead — NSF AI Institute Arena on AI for Interactive Coaches and Services	2023–2024
Virtual Experience Chair — Conference on Robot Learning (CoRL)	2023
Seminar and Panel Organizer , Life as a Professor, CMU	2023
Semesterly seminar where a panel of CMU faculty discuss their perspectives on becoming and being a professor for graduate students interested in the academic career path	

Special Session Organizer — RO-MAN, Special Session on Human-Agent/Robot Interaction in Health-care and Medicine	2023
Workshop Organizer — ICRA, Emerging paradigms for assistive robotic manipulation: from research labs to the real world	2023
PhD Admissions Committee — CMU, Robotics Institute	2022–
Chair — IROS 2022 session on Art and Entertainment and Manipulation	2022
Co-Chair — ICRA 2022 session on Physical HRI	2022
Area Chair — Conference on Robot Learning (CoRL)	2022–2023
Workshop Organizer — IROS, 5th Workshop on Proximity Perception - Towards Next-Generation Multi-Modal Sensing in Soft Structures	2022
Associate Editor IEEE Conference on Robot and Human Interactive Communication (RO-MAN)	2021, 2022
Reviewer T-RO, ICRA, IROS, RA-L, CoRL, CVPR, ICML, ICLR, NeurIPS, Humanoids, HRI, Sensors, RO-MAN, Science Robotics, RSS Pioneers	
Workshop Organizer — ICRA, Learning for Caregiving Robots	2021
Seminar Organizer , Georgia Tech Life as a Professor: Student Advising and Recruiting	2020
Life as a Professor: Funding 101	2019
Life as a Professor: Starting a Start-Up	2018
Panelist Graduate Intro to Robotics Research, Georgia Tech	2018
Summer Undergraduate Research Experience (SURE) Program, Georgia Tech	2017–2019
RoboGrads (robotics graduate student organization), Georgia Tech Vice President for Robotics PhD	2019–2020
President	2018–2019
Student Thesis Committees Vihaan Misra, CMU, MSR	2025
Caleb Escobedo, CU Boulder, PhD	2024–
Christian Berger, CMU, MSR	2024
Christopher Chang, CMU, MSR	2024
Zhiyang (Jerry) He, UC Berkeley, PhD	2023–
Student Qualifying Exam Committees Maggie Collier, CMU, PhD	2024
Ruixuan Liu, CMU, PhD	2023

Outreach

CEO Soak , ALS Association, Pittsburgh Organized a table showcasing robotics research at CMU that can impact individuals with ALS.	2025
AI Scholars , Co-organizer, CMU Each summer, AIS provides 30+ rising high school seniors who have historically been excluded from STEM fields the opportunity to explore artificial intelligence through 4 weeks of courses and hands-on projects with Carnegie Mellon faculty, staff, and researchers.	2023–

AI4ALL , Co-organizer, CMU	2022
RoboGrads , Vice President of Outreach, Georgia Tech Organized K-12 outreach events for over 30 robotics labs at Georgia Tech	2017–2018
Biomedical Robotics Club , Mentor, Georgia Tech Mentoring over 50 undergraduate students in how to research and build assistive devices and robots for people with impairments.	2016–2019
FIRST Lego League , Judge	2015–2021
CS Outreach & Diversity Club , UW–La Crosse Organizing CS and robotics events for K-12 students.	2015–2016
FIRST Robotics , Mentor, Central High School and Holmen High School	2012–2016

Peer-Reviewed Publications (Conferences and Journals)

- [1] **Bidirectional Human-Robot Communication for Physical Human-Robot Interaction**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2026 (**Best Paper Finalist**)
Junxiang Wang, Cindy Wang, Rana Soltani Zarrin, Zackory Erickson
- [2] **WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2026 (**Best Paper Award**)
Akhil Padmanabha*, Jessie Yuan*, Tanisha Mehta, Rajat Kumar Jenamani, Eric Hu, Victoria de León, Anthony Wertz, Janavi Gupta, Ben Dodson, Yunting Yan, Carmel Majidi, Tapomayukh Bhattacharjee[†], Zackory Erickson[†]
- [3] **Knowing When to Help: A Framework for Error Detection in Meal Preparation Tasks for Older Adults**
ACM/IEEE International Conference on Human-Robot Interaction (HRI) Late Breaking Reports, 2026
Michaela Tecson, Zackory Erickson, and Reid Simmons
- [4] **AI in geriatric psychiatry: precision meets human experience**
Neuropsychopharmacology, 2026
Akiko Mizuno, Zackory Erickson, Daniel E. Jimenez, and Howard Jay Aizenstein
- [5] **In Vivo Skin 3-D Surface Reconstruction and Wrinkle Depth Estimation Using Handheld High Resolution Tactile Sensing**
Advanced Healthcare Materials, 2025
Akhil Padmanabha, Arpit Agarwal, Catherine Li, Austin Williams, Dinesh K Patel, Sankalp Chopkar, Achu Wilson, Ahmet Ozkan, Wenzhen Yuan, Sonal Choudhary, Arash Mostaghimi, Zackory Erickson, Carmel Majidi
- [6] **Geometric Red-Teaming for Robotic Manipulation**
Conference on Robot Learning (CoRL), 2025 (**Oral Presentation**)
Divyam Goel, Yufei Wang, Tiancheng Wu, Helen Qiao, Pavel Piliptchak, David Held[†], Zackory Erickson[†]
- [7] **Force-Modulated Visual Policy for Robot-Assisted Dressing with Arm Motions**
Conference on Robot Learning (CoRL), 2025
Alexis Yihong Hao, Yufei Wang, Navin Sriram Ravie, Bharath Hegde, David Held[†], Zackory Erickson[†]
- [8] **CoRI: Synthesizing Communication of Robot Intent for Physical Human-Robot Interaction**
Conference on Robot Learning (CoRL), 2025
Junxiang Wang, Emek Barış Küçükatabak, Rana Soltani Zarrin, Zackory Erickson
- [9] **ORB: Operating Room Bot, Automating Operating Room Logistics through Mobile Manipulation**
International Conference on Automation Science and Engineering (CASE), 2025

Jinkai Qiu*, Yungjun Kim*, Gaurav Sethia*, Tanmay Agarwal*, Siddharth Ghodasara*, Zackory Erickson, Jeffrey Ichnowski

- [10] **Leveraging Large Language Models for Preference-Based Sequence Prediction**
International Conference on Agents and Artificial Intelligence (ICAART 2025), 2025
Michaela Tecson, Daphne Chen, Michelle Zhao, Zackory Erickson, Reid Simmons
- [11] **Incremental Learning for Robot Shared Autonomy**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Yiran Tao, Guixiu Qiao, Dan Ding, Zackory Erickson
- [12] **Real-World Offline Reinforcement Learning from Vision Language Model Feedback**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Sreyas Venkataraman*, Yufei Wang*, Ziyu Wang, Navin Sriram Ravie, Zackory Erickson[†], David Held[†]
- [13] **SkinGrip: An Adaptive Soft Robotic Manipulator with Capacitive Sensing for Whole-Limb Bed Bathing Assistance**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Fukang Liu, Kavya Puthuveetil, Akhil Padmanabha, Karan Khokar, Zeynep Temel, Zackory Erickson
- [14] **DiSRT-In-Bed: Diffusion-Based Sim-to-Real Transfer Framework for In-Bed Human Mesh Recovery**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
Jing Gao, Ce Zheng, Laszlo A Jeni, Zackory Erickson
- [15] **RoboCAP: Robotic Classification and Precision Pouring of Diverse Liquids and Granular Media with Capacitive Sensing**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Yexin Hu*, Alexandra Gillespie*, Akhil Padmanabha, Kavya Puthuveetil, Wesley Lewis, Karan Khokar, Zackory Erickson
- [16] **ArticuBot: Learning Universal Articulated Object Manipulation Policy via Large Scale Simulation**
Robotics Science and Systems (RSS), 2025
Yufei Wang*, Ziyu Wang*, Mino Nakura, Pratik Bhowal, Chia-Liang Kuo, Yi-Ting Chen, Zackory Erickson[†], David Held[†]
- [17] **High-density Electromyography for Effective Gesture-based Control of Physically Assistive Mobile Manipulators**
npj Robotics, 2025
Jehan Yang, Kent Shibata, Douglas Weber, Zackory Erickson
- [18] **LAMS: LLM-Driven Automatic Mode Switching for Assistive Teleoperation**
ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025 (Best Paper finalist)
Yiran Tao*, Jehan Yang*, Dan Ding, Zackory Erickson
- [19] **EMGBench: Benchmarking Out-of-Distribution Generalization and Adaptation for Electromyography**
Advances in neural information processing systems (NeurIPS), 2024
Jehan Yang, Maxwell Soh, Vivianna Lieu, Douglas J Weber, Zackory Erickson
- [20] **VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots**
ACM Symposium on User Interface Software and Technology (UIST), 2024
Akhil Padmanabha*, Jessie Yuan*, Janavi Gupta, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, Zackory Erickson
- [21] **DiffTORI: Differentiable Trajectory Optimization for Deep Reinforcement and Imitation Learning**
Advances in neural information processing systems (NeurIPS), 2024 (Spotlight Presentation)
Weikang Wan*, Ziyu Wang*, Yufei Wang*, Zackory Erickson[†], David Held[†]

- [22] **RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation**
International Conference on Machine Learning (ICML), 2024
 Yufei Wang*, Zhou Xian*, Feng Chen*, Tsun-Hsuan Wang, Yian Wang, Katerina Fragkiadaki, Zackory Erickson, David Held, Chuang Gan
- [23] **RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback**
International Conference on Machine Learning (ICML), 2024
 Yufei Wang*, Zhanyi Sun*, Jesse Zhang, Zhou Xian, Erdem Bıyık, David Held†, Zackory Erickson†
- [24] **Do Mistakes Matter? Comparing Trust Responses of Different Age Groups to Errors Made by Physically Assistive Robots**
IEEE International Conference on Robot and Human Interactive Communication (ROMAN), 2024
 Sasha Wald*, Kavya Puthuveetil*, Zackory Erickson
- [25] **Unfolding the literature: A review of robotic cloth manipulation**
Annual Review of Control, Robotics, and Autonomous Systems, 2024
 Alberta Longhini, Yufei Wang, Irene Garcia-Camacho, David Blanco-Mulero, Marco Moletta, Michael Welle, Guillem Alenyà, Hang Yin, Zackory Erickson, David Held, Júlia Borràs, Danica Kragic
- [26] **AdaFold: Adapting Folding Trajectories of Cloths via Feedback-Loop Manipulation**
IEEE Robotics and Automation Letters (RA-L), 2024
 Alberta Longhini, Michael C. Welle, Zackory Erickson, Danica Kragic
- [27] **BodyMAP-Jointly Predicting Body Mesh and 3D Applied Pressure Map for People in Bed**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
 Abhishek Tandon*, Anujraaj Goyal*, Henry M Clever, Zackory Erickson
- [28] **Force Constrained Visual Policy: Safe Robot-Assisted Dressing via Multi-Modal Sensing**
IEEE Robotics and Automation Letter (RA-L), 2024
 Zhanyi Sun*, Yufei Wang*, David Held†, Zackory Erickson†
- [29] **Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator**
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2024 (**Best Paper Award**)
 Akhil Padmanabha, Janavi Gupta, Chen Chen, Jehan Yang, Vy Nguyen, Douglas J Weber, Carmel Majidi, Zackory Erickson
- [30] **Quantifying Assistive Robustness Via the Natural-Adversarial Frontier**
Conference on Robot Learning (CoRL), 2023
 Jerry Zhi-Yang He, Daniel S Brown, Zackory Erickson, Anca Dragan
- [31] **A Multimodal Sensing Ring for Quantification of Scratch Intensity**
Nature Communications Medicine, 2023
 Akhil Padmanabha, Sonal Choudhary, Carmel Majidi, Zackory Erickson
- [32] **Robust Body Exposure (RoBE): A Graph-based Dynamics Modeling Approach to Manipulating Blankets over People**
IEEE Robotics and Automation Letters (RA-L), 2023
 Kavya Puthuveetil, Sasha Wald, Atharva Pusalkar, Pratyusha Karnati, and Zackory Erickson
- [33] **One Policy to Dress Them All: Learning to Dress People with Diverse Poses and Garments**
Robotics Science and Systems (RSS), 2023
 Yufei Wang, Zhanyi Sun, Zackory Erickson†, David Held†
- [34] **HAT: Head-Worn Assistive Teleoperation of Mobile Manipulators**
IEEE International Conference on Robotics and Automation (ICRA), 2023
 Akhil Padmanabha*, Qin Wang*, Daphne Han, Jashkumar Diyora, Kriti Kacker, Hamza Khalid, Liang-Jung Chen, Carmel Majidi, Zackory Erickson

- [35] **SLURP! Spectroscopy of Liquids Using Robot Pre-Touch Sensing**
IEEE International Conference on Robotics and Automation (ICRA), 2023
 Nathaniel Hanson*, Wesley Lewis*, Kavya Puthuveetil, Donelle Furline Jr, Akhil Padmanabha, Taskin Padir, Zackory Erickson
- [36] **EDO-Net: Learning Elastic Properties of Deformable Objects from Graph Dynamics**
IEEE International Conference on Robotics and Automation (ICRA), 2023
 Alberta Longhini*, Marco Moletta*, Alfredo Reichlin, Michael C. Welle, David Held, Zackory Erickson, Danica Kragic
- [37] **Elastic Context: Encoding Elasticity for Data-driven Models of Textiles**
IEEE International Conference on Robotics and Automation (ICRA), 2023
 Alberta Longhini, Marco Moletta, Alfredo Reichlin, Michael C. Welle, Alexander Kravberg, Yufei Wang, David Held, Zackory Erickson, Danica Kragic
- [38] **A Study of Causal Confusion in Preference-Based Reward Learning**
International Conference on Learning Representations (ICLR), 2023
 Jeremy Tien, Jerry Zhi-Yang He, Zackory Erickson, Anca D. Dragan, Daniel S. Brown
- [39] **ToolFlowNet: Robotic Manipulation with Tools via Predicting Tool Flow from Point Clouds**
Conference on Robot Learning (CoRL), 2022
 Daniel Seita, Yufei Wang, Edward Yao Li, Sarthak J Shetty, Zackory Erickson, and David Held
- [40] **Learning Representations that Enable Generalization in Assistive Tasks**
Conference on Robot Learning (CoRL), 2022
 Jerry Zhi-Yang He, Zackory Erickson, Daniel S. Brown, Aditi Raghunathan, and Anca Dragan
- [41] **Characterization of a Meso-Scale Wearable Robot for Bathing Assistance**
IEEE International Conference on Robotics and Biomimetics (ROBIO), 2022
 Fukang Liu, Vaidehi Patil, Zackory Erickson, and Zeynep Temel
- [42] **Visual Haptic Reasoning: Estimating Contact Forces by Observing Deformable Object Interactions**
IEEE Robotics and Automation Letters (RA-L), 2022
 Yufei Wang, David Held, and Zackory Erickson
- [43] **CapSense: A Real-Time Capacitive Sensor Simulation Framework for Physical Human-Robot Interaction**
IEEE Robotics and Automation Letters (RA-L), 2022
 Christian Schöffmann, Zackory Erickson, and Hubert Zangl
- [44] **Bodies Uncovered: Learning to Manipulate Real Blankets Around People via Physics Simulations**
IEEE Robotics and Automation Letters (RA-L), 2022
 Kavya Puthuveetil, Charles C. Kemp, and Zackory Erickson
- [45] **Characterizing Multidimensional Capacitive Servoing for Physical Human-Robot Interaction**
IEEE Transactions on Robotics (T-RO), 2022
 Zackory Erickson, Henry M. Clever, Vamsee Gangaram, Eliot Xing, Greg Turk, C. Karen Liu, and Charles C. Kemp
- [46] **Assistive VR Gym: Interactions with Real People to Improve Virtual Assistive Robots**
IEEE Conference on Robot and Human Interactive Communication (RO-MAN), 2020
 Zackory Erickson*, Yijun Gu*, and Charles C. Kemp
- [47] **Multimodal Material Classification for Robots using Spectroscopy and High Resolution Texture Imaging**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020
 Zackory Erickson, Eliot Xing, Bharat Srirangam, Sonia Chernova, and Charles C. Kemp

- [48] **Bodies at Rest: 3D Human Pose and Shape Estimation from a Pressure Image using Synthetic Data**
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020 (**Oral**)
 Henry M. Clever, Zackory Erickson, Ariel Kapusta, Greg Turk, C. Karen Liu, and Charles C. Kemp
- [49] **Assistive Gym: A Physics Simulation Framework for Assistive Robotics**
IEEE International Conference on Robotics and Automation (ICRA), 2020
 Zackory Erickson, Vamsee Gangaram, Ariel Kapusta, C. Karen Liu, and Charles C. Kemp
- [50] **Learning to Collaborate from Simulation for Robot-Assisted Dressing**
IEEE Robotics and Automation Letters (RA-L), 2020
 Alexander Clegg, Zackory Erickson, Patrick Grady, Greg Turk, Charles C. Kemp, and C. Karen Liu
- [51] **Active Robot-Assisted Feeding with a General-Purpose Mobile Manipulator: Design, Evaluation, and Lessons Learned**
Robotics and Autonomous Systems, 2020
 Daehyung Park, Yuuna Hoshi, Harshal P. Mahajan, Ho Keun Kim, Zackory Erickson, Wendy A. Rogers, Charles C. Kemp
- [52] **Multidimensional Capacitive Sensing for Robot-Assisted Dressing and Bathing**
IEEE International Conference on Rehabilitation Robotics (ICORR), 2019 (**Best Student Paper**)
 Zackory Erickson, Henry M. Clever, Vamsee Gangaram, Greg Turk, C. Karen Liu, and Charles C. Kemp
- [53] **Classification of Household Materials via Spectroscopy**
IEEE Robotics and Automation Letters (RA-L), 2019 (**Best Paper Award in Service Robotics Finalist at ICRA 2019**)
 Zackory Erickson, Nathan Luskey, Sonia Chernova, and Charles C. Kemp
- [54] **Personalized Collaborative Plans for Robot-Assisted Dressing via Optimization and Simulation**
Autonomous Robots, 2019
 Ariel Kapusta, Zackory Erickson, Henry M. Clever, Wenhao Yu, C. Karen Liu, Greg Turk, and Charles C. Kemp
- [55] **Autonomous Tool Construction Using Part Shape and Attachment Prediction**
Robotics: Science and Systems (RSS), 2019
 Lakshmi Nair, Nithin Srikanth, Zackory Erickson, Sonia Chernova
- [56] **Deep Haptic Model Predictive Control for Robot-Assisted Dressing**
IEEE International Conference on Robotics and Automation (ICRA), 2018
 Zackory Erickson, Henry M. Clever, Greg Turk, C. Karen Liu, and Charles C. Kemp
- [57] **Tracking Human Pose During Robot-Assisted Dressing using Single-Axis Capacitive Proximity Sensing**
IEEE Robotics and Automation Letters (RA-L), 2018
 Zackory Erickson, Maggie Collier, Ariel Kapusta, and Charles C. Kemp
- [58] **3D Human Pose Estimation on a Configurable Bed from a Pressure Image**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2018
 Henry M. Clever, Ariel Kapusta, Daehyung Park, Zackory Erickson, Yash Chitalia, Charles C. Kemp
- [59] **Semi-Supervised Haptic Material Recognition for Robots using Generative Adversarial Networks**
Conference on Robot Learning (CoRL), 2017
 Zackory Erickson, Sonia Chernova, and Charles C. Kemp
- [60] **What Does the Person Feel? Learning to Infer Applied Forces During Robot-Assisted Dressing**
IEEE International Conference on Robotics and Automation (ICRA), 2017
 Zackory Erickson, Alexander Clegg, Wenhao Yu, Greg Turk, C. Karen Liu, and Charles C. Kemp

- [61] **Learning to Navigate Cloth using Haptics**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
 Alexander Clegg, Wenhao Yu, Zackory Erickson, C. Karen Liu, and Greg Turk
- [62] **A Multimodal Execution Monitor with Anomaly Classification for Robot-Assisted Feeding**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017
 Daehyung Park, Hokeun Kim, Yuuna Hoshi, Zackory Erickson, Ariel Kapusta, and Charles C. Kemp
- [63] **Multimodal Execution Monitoring for Anomaly Detection During Robot Manipulation**
IEEE International Conference on Robotics and Automation (ICRA), 2016
 Daehyung Park, Zackory Erickson, Tapomayukh Bhattacharjee, and Charles C. Kemp

Workshop Papers

- [1] **Bimanual Wearable EMG Control for Multi-Day In-Home Mobile Manipulation by a User with Quadriplegia**
HRI 2026: Workshop on Lifelong Learning and Personalization in Long-Term Human-Robot Interaction, 2026
 Jehan Yang, Eleanor Hodgson, Cindy Sun, Zackory Erickson and Douglas J. Weber
- [2] **RaC: Robot Learning for Long-Horizon Tasks by Scaling Recovery and Correction**
CoRL 2025: Workshop on Making Sense of Data in Robotics: Composition, Curation, and Interpretability at Scale, 2025
 Zheyuan Hu, Robyn Wu, Naveen Enock, Jasmine Jia-ni Li, Riya Kadakia, Zackory Erickson, Aviral Kumar
- [3] **Towards Wearable Interfaces for Robotic Caregiving**
HRI 2025: Workshop on Physical Caregiving Robots, 2025
 Akhil Padmanabha, Carmel Majidi, Zackory Erickson
- [4] **Towards an LLM-Based Speech Interface for Robot-Assisted Feeding**
UIST 2024 demo track: Symposium on User Interface Software and Technology, 2024
 Jessie Yuan, Janavi Gupta, Akhil Padmanabha, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, Zackory Erickson
- [5] **Towards the Development of Wound Care Robots: An Observational Study Outlining Robotic Design Needs for Wound Care**
HRI 2024: Workshop on HRI for Aging in Place, 2024
 Zulekha Karachiwalla, Ellen Lee, Andrew Dierkes, Zackory Erickson, Henny Admoni
- [6] **Elastic Context: Encoding Elasticity for Data-driven Models of Textiles**
ICRA 2023: Embracing contacts. Making robots physically interact with our world, 2023
 Alberta Longhini, Marco Moletta, Alfredo Reichlin, Michael C. Welle, Alexander Kravberg, Yufei Wang, David Held, Zackory Erickson, Danica Kragic
- [7] **HAT: Head-Worn Assistive Teleoperation of Mobile Manipulators**
ICRA 2023: Emerging paradigms for assistive robotic manipulation: from research labs to the real world, 2023
 Akhil Padmanabha*, Qin Wang*, Daphne Han, Jashkumar Diyora, Kriti Kacker, Hamza Khalid, Liang-Jung Chen, Carmel Majidi, Zackory Erickson
- [8] **A Study of Causal Confusion in Preference-Based Reward Learning**
ICML 2022: Workshop on Spurious Correlations, Invariance and Stability, 2022
 Jeremy Tien, Jerry Zhi-Yang He, Zackory Erickson, Anca Dragan, Daniel S. Brown